DRUGS · ANTIBIOTICS: OTHER CLASSES

Metronidazole

Metronidazole is a nitroimidazole that kills anaerobic bacteria and some protozoa. In head and neck practice it is the anaerobic partner added to a penicillin or cephalosporin for dental, peritonsillar, deep neck, otogenic and sinogenic intracranial infection, and for prophylaxis when the pharynx is opened. Its classic warnings are alcohol and warfarin.

1 CHAPTER ONE At a glance

The drug on one page.

1.1 Summary

METRONIDAZOLE

	SUMMARY
Class	Nitroimidazole
How it works	Reduced inside anaerobic cells to reactive nitro radicals that break DNA; bactericidal
Spectrum	Obligate anaerobes (Bacteroides, Prevotella, Fusobacterium, Peptostreptococcus, Clostridia), protozoa (Trichomonas, Giardia, Entamoeba); no activity against aerobes or Actinomyces
ENT uses	With a beta-lactam for quinsy, deep neck infection, dental abscess, Lemierre syndrome, otogenic and sinogenic intracranial abscess, chronic and complicated sinusitis, necrotising ulcerative gingivitis; head and neck surgical prophylaxis; C. difficile (oral) in mild disease where vancomycin is not available
Adult dose	Oral 400 mg three times daily (or 500 mg three times daily); intravenous 500 mg every 8 hours; rectal 1 g every 8 hours
Child dose	7.5 mg/kg three times daily orally or intravenously (maximum 500 mg)
Avoid	Alcohol during and for 48 hours after; caution in liver disease

	SUMMARY
Pregnancy	Avoid high-dose regimens; normal doses used when needed

2 CHAPTER TWO How it works

Mechanism and handling.

2.1 Mechanism

A prodrug activated only in the low-redox environment of anaerobes, which is why it is selective. Resistance is rare.

2.2 Handling

Almost completely absorbed orally (so intravenous use is only for patients who cannot swallow), penetrates abscesses and the brain very well, metabolised in the liver.

3 CHAPTER THREE Using it in ENT

Indications and doses.

3.1 Mixed infections of the head and neck

Odontogenic, peritonsillar, parapharyngeal and retropharyngeal infections and **Lemierre syndrome** (Fusobacterium necrophorum thrombophlebitis of the internal jugular vein) are mixed or anaerobic: give a penicillin or cephalosporin with metronidazole, and drain collections.

3.2 Intracranial complications

For otogenic or sinogenic **brain abscess**, subdural empyema and venous sinus thrombosis, metronidazole with high-dose ceftriaxone is the standard empirical combination because of its excellent brain penetration.

3.3 Prophylaxis

Given with cefuroxime or co-amoxiclav at induction for clean-contaminated surgery such as laryngectomy and oral cavity resection.

4 CHAPTER FOUR Safety

What goes wrong.

4.1 Adverse effects

- Metallic taste, nausea, furred tongue, dark urine.
- **Disulfiram-like reaction** with alcohol: flushing, vomiting and tachycardia.
- **Peripheral and optic neuropathy** and encephalopathy with prolonged courses (beyond about 10 days or high totals); seizures.
- Hepatotoxicity, severe in Cockayne syndrome.

4.2 Interactions

Warfarin (marked rise in INR), alcohol, lithium (toxicity), phenytoin and busulfan; phenobarbital lowers its levels.

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Selectivity for anaerobes and why that is.
- Organisms of **Lemierre syndrome** and its treatment.
- The **alcohol** and **warfarin** interactions; neuropathy with long courses.

5.2 Common pitfalls

COMMON PITFALLS

- Giving intravenous metronidazole to a patient who can swallow.
- Forgetting to warn about alcohol.
- Long courses without review for neuropathy.

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